

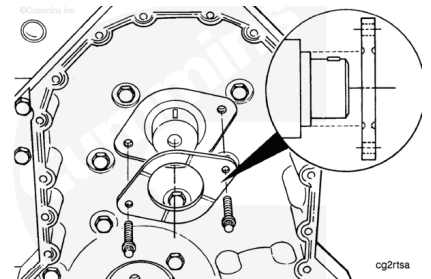
Trainings ▾

Install

Install the thrust plate on the camshaft.

Install the capscrews and tighten.

Torque Value: 47 n•m [35 ft-lb]

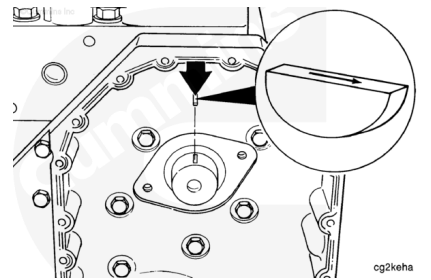


Gear housing clearance makes key installation more difficult.
Use care in seating the key.

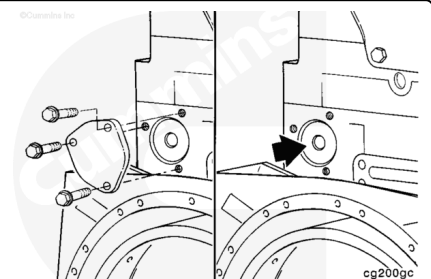
Be sure the gear mounting surface is clean and free of oil, dirt, or debris.

If the camshaft gear is used again, use the same part number key as the one that was removed. Be sure the arrow on the key is pointing in the same direction as when it was removed.

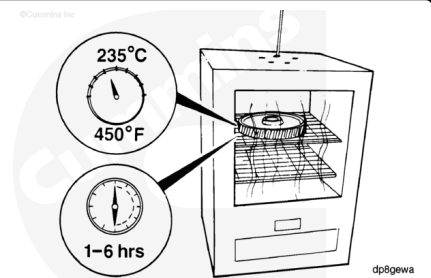
Use a leather hammer to install the camshaft gear key.



It can be necessary to remove the camshaft rear cover plate and apply pressure to the end of the camshaft to hold it in the forward position while the gear is being installed. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html>).



Heat the gear in an oven for a minimum of one hour, but **not** longer than six hours.

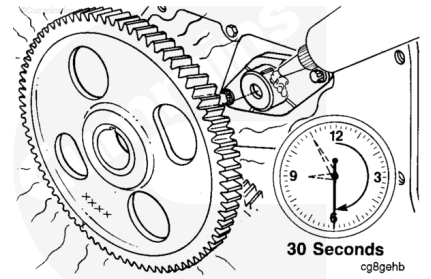


Measurements

	celsius	fahrenheit
Temperature:	235	450

⚠ WARNING ⚠

Use insulated gloves, Part Number 3823730, and/or hot clamp pliers, Part Number 3823732, when handling heated parts. Hot parts can cause serious personal injury.



⚠ CAUTION ⚠

The timing marks and gear part number must be facing away from the camshaft when the gear is installed to prevent engine damage.

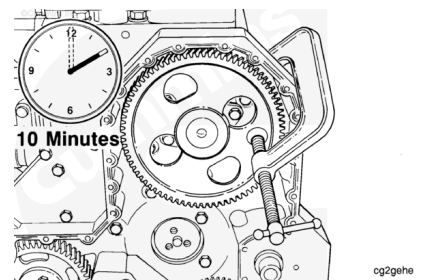
Use Lubriplate™ 105, or equivalent, to coat the camshaft nose before installing the camshaft gear.

Remove the gear from the oven. Install the gear on the camshaft within 30 seconds after removing from the oven.

Align the gear keyway with the key in the camshaft, and install the gear on the camshaft.

Install a C-clamp around the gear housing and the camshaft gear to hold the gear in position as it cools.

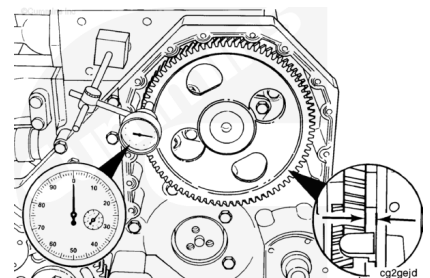
Remove the C-clamp after the gear has cooled to room temperature (approximately 10 minutes).



Use a dial indicator to check the end clearance of the gear.

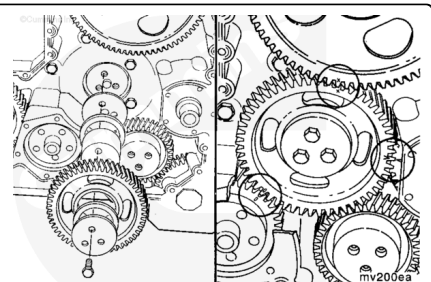
Push the camshaft to the rear and set the gauge at zero.

Pull the gear out and measure the end clearance.



mm		in
0.13	MIN	0.005
0.41	MAX	0.016

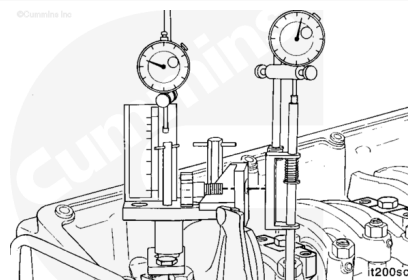
Install the camshaft idler gear. Refer to Procedure 001-036 (</qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html>).



⚠ CAUTION ⚠

Failure to check the engine timing can cause severe engine damage.

Check the engine timing. Refer to Procedure 006-025
(/qs3/pubsys2/xml/en/procedures/35/35-006-025-tr.html).



Install the gear cover. Refer to Procedure 001-031
(/qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html).

Operate the engine to normal operating temperature and check for leaks.

